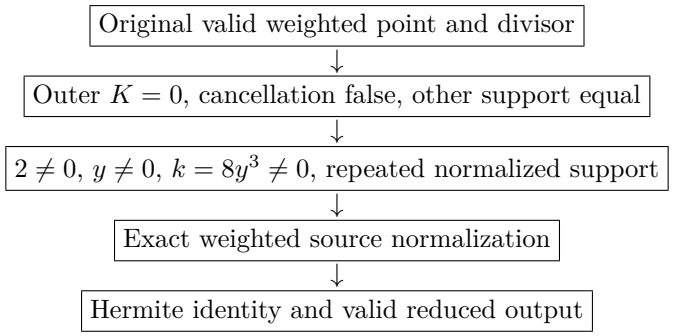
Repeated-support tripling: weighted field branch

Result. Six compiled declarations connect the source repeated-support guards to the exact six-field tripling output and normalized Mumford validity. The final theorem uses original point/divisor validity and source guards; it assumes no intermediate or output validity and no nonzero derivative.



Assumptions. An arbitrary field, the monic quintic $f(T) = T^5 + a_3T^3 + a_2T^2 + a_1T + a_0$, original normalized degree-one and degree-two divisibility, four nonzero representation weights, outer raw $K = 0$, cancellation comparison false, and equal cross-scaled other coordinate. The positive ordinate alternative is derived. The equal-coordinate condition gives normalized support $(T - x)^2$; cancellation failure excludes characteristic two and zero original ordinate.

Construction. Put $m = f'(x)/(2y)$ and $A = (10x^3 + 3a_3x + a_2 - m^2)/(2y)$. Define $h(T) = A(T - x)^2 + m(T - x) + y$, $q = 3x - A^2$, and $r = a_3 - 2A(m - 2Ax) + 3x(q - x)$. The checked polynomial identity is

$$f - h^2 = (T - x)^3(T^2 + qT + r),$$

under the original point equation. The output ordinate is

$$v = (Aq - (m - 2Ax))T + Ar - (y - mx + Ax^2).$$

Since $h + v = A(T^2 + qT + r)$, the reduced support divides $f - v^2$.

Source correspondence. rawOutput reproduces triple lines285–345 of the retained Rust source, including its fp, fpp, k, aa, bb, cc and six returned fields. raw_normalizes proves their weighted normalization equals the construction above and that both returned weights are nonzero. source_repeated_preserves connects this to mixed dispatcher lines828–878. The source returns a degree-two result unconditionally on this selected path.

Validation. Main attempt4 and separate readback pass; all six declarations use only propext, Classical.choice and Quot.sound. Full nonunit rational fixtures invoke the original-input theorem for $f = T^5 - T + 1$ and $f = T^5 + 1$ at normalized $(0, 1)$ with repeated support. The second has zero first and second derivatives and produces normalized $(q, r, s, t) = (0, 0, 0, -1)$. Additional controls check zero-ordinate cancellation and invalid original-point rejection. Receipts and exact source hashes accompany the report.

Preserved attempts and boundary. The first proof attempt lacked a noncomputable annotation and used an over-specific zero rewrite. Fixture attempts retained rational-coefficient simplification failures before explicit polynomial coefficient identities were supplied. Literal UInt64 correspondence remains a next component: constants8 and10 need canonical-residue reduction and no-overflow bounds, especially at p=5 and p=7. This packet proves field values, not counters, compiler execution, divisor-class addition or complete walker closure. No measurement or scientific trial was run; cost ratio is null. Independent source review follows sealing.